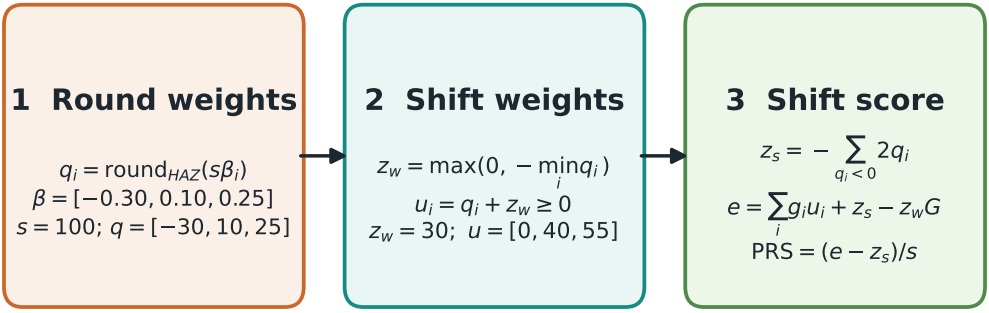


Converting signed decimal weights to nonnegative integers

Weights are rounded and shifted; decoding restores the score after calculation



Worked example begins with Equation 1

$g = [0, 2, 1]$; PRS from Equation 1 = 0.45
shifted weighted sum = 135; $G = 3$; $z_s = 60$
encoded $e = 105$; decoded PRS = 0.45

Safety condition for all values used in the calculation: $4sMN \leq 2^{64} - 1$